SAMPLE

Earth's Sky: Light Pollution

How Artificial Light Hides the Stars

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Introduction

For most of the history of the human race, the night was dark. The Sun set, the stars came out, and the sky overhead was a presence as constant as the land: the Milky Way arching from horizon to horizon, the planets tracking their slow courses among fixed stars, the darkness itself a fact of life that every culture organized itself around. That sky is gone for most of the species. The modern night, over the cities and suburbs where nearly everyone lives, carries a glow of the world's own making, the light of a hundred billion lamps scattered back down by the air, and the stars that our ancestors took for granted have become a rare sight, a thing people travel to see. The glow has a name, light pollution, and the story of its rise, its costs, and the growing movement to undo it is the subject of this book.

Light pollution is not one thing but several. The glow that washes out the stars over the city is skyglow, the light sent upward by outdoor lamps and scattered back to the ground by molecules and aerosols in the atmosphere; the harsh brilliance of a unshielded fixture in the eye is glare; the neighbor's porch light falling across a bedroom window is light trespass; the visual chaos of overlit streets and competing signs is clutter. The four kinds share a cause, the design of the modern lighting estate, and they share a remedy, the design of better lighting, which is why the subject, though born in astronomy's complaint about the stars, has grown into a general environmental science, with consequences tracked through ecology, human health, energy policy, and the culture of the night itself.

The subject's science is real and deep. The physics of skyglow is the physics of light scattering in air, the same machinery that makes the daytime sky blue; the biology is the biology of circadian rhythm, the ancient light-driven clock that every animal body carries, from insects to humans, and the growing evidence of what that clock suffers under constant light; the ecology is the ecology of a world of nocturnal creatures, thousands of species evolved for the dark, now living under a permanent twilight that changes their feeding, their breeding, and their deaths; and the measurement side has produced its own instruments and standards, from the naked-eye star counts of amateur networks to the satellite maps of the planet's night side and the sky-quality meters that let anyone put a number on their own night.

A note on the book's scope and its numbers, before the story starts. The subject, as the chapters that follow treat it, runs from the physics of the scattered photon to the politics of the streetlight contract, and the account draws on the field's whole literature: the atmospheric science of the glow, the astronomy of the brightened sky, the ecology of the lit night, the medicine of the circadian rhythm, and the practice of the lighting craft. The numbers quoted are the field's established findings, the atlas fractions and the record's growth rates, stated as the literature states them; the case studies, the cities and the parks, are the movement's documented examples; and where the science is unsettled, the insect decline's causes, the safety statistics' interpretations, the account says so. The subject's own honest rule, learned over its century, is the one this book keeps: the night deserves the same care in its telling as the science gives it in its measuring, and the story of the world's most recoverable environmental loss is best told exactly as it is.

This book follows the subject in full: the discovery of the problem, as astronomers watched the observatories' skies brighten and the cities' glow spread; the science of what the light does, to the sky, to the animals, and to us; the measurement of the night, by eye, by meter, and by satellite; the stories of the places that acted, the dark-sky cities and reserves that now protect some of the last true nights; and the solutions, the lighting craft that can keep the streets safe and the stars visible, and the movement that is teaching it to the world. The night was the first commons, the sky that belonged to everyone, and the story of its loss and recovery is, in the end, a story about what a civilization wants from its own lights.

Chapter 1: The Discovery of Light Pollution

The discovery of light pollution, as a named problem, belongs to the astronomers, because the astronomers were the first to lose something they could measure. The great observatories of the late nineteenth and early twentieth centuries were built on hills near cities, Mount Wilson above Los Angeles, the telescopes of the American and European institutes within sight of town, chosen for their clear air and dark nights; and through the century's middle decades the cities grew, and the glow came up the hills. The astronomers' complaint, at first a matter of professional inconvenience, hardened into a science: measurements of the sky's brightening were published, the physics of the glow was worked out, and by the 1970s the term light pollution had entered the literature, an entry in astronomy's long ledger of the ways the ground interferes with the sky.

The popular discovery came through a different door, the door of the lost Milky Way. The great sky of the human inheritance faded gradually enough that its loss was felt before it was named: city dwellers of the mid-century spoke of seeing the stars from town in their childhoods, of the Milky Way visible before the streetlights came, and the generation that noticed the change found no science to name it, only the general sense that the modern night was brighter than it ought to be. The name arrived in the public vocabulary when the astronomers' literature reached the magazines, and the phrase caught at once, because it named a thing everyone had half-noticed: a night so bright that the sky, the great sky, was simply missing.

The institutional discovery was the founding of the movement. In 1988, two astronomers, David Crawford and Timothy Hunter, founded the International Dark-Sky Association in Arizona, an organization dedicated to the protection of the night sky, and the IDA's program became the movement's framework: the science of good lighting, the catalog of fixtures that light the ground instead of the sky, the advocacy for ordinances and codes, and the recognition, through its dark-sky places programs, of the parks and reserves and communities that protect their nights. The association's founding marks the subject's coming of age: the astronomers' private complaint had become an organized public cause, with a literature, a legal program, and a growing membership across the world.

The scientific discovery, in the same decades, came through the biology. The light at night, once assumed to be a matter of astronomy and aesthetics alone, was found in the 1980s and 1990s to be a biological force of the first order: the circadian rhythm's sensitivity to night light was measured, the hormone melatonin's suppression by light was documented, and the ecological literature filled with case after case, from the sea turtles of the Florida beaches to the migrating birds of the lit towers, of species living wrongly under the electric twilight. The biological discovery transformed the subject: the astronomers' problem, the wash of the stars, was revealed as the visible edge of a general environmental change, the remaking of the night for every living thing, and the science of light pollution took its modern shape, an interdisciplinary study with a shared cause and a shared remedy.

A further thread of the discovery belongs in the chapter, the amateur astronomers' part, because the subject's first data came from the backyard. The amateur community, watching its own skies brighten through the century's middle decades, kept the first records: the visual estimates of the limiting magnitude, the counts of the visible stars, the comparisons of the club's observing field year by year, all collected in the societies' journals and newsletters long before the professional literature carried the subject. The amateurs' testimony, assembled across the countries, gave the movement its early evidence and its early constituency, and the discovery's institutional story, the association's founding in 1988, is the record of the amateurs' complaint finding its organization: the backyard observers, the planetarium staff, and the professional astronomers of the darkening sites, joining into the body that gave the problem its name in the public's ear.

A closing item for the discovery chapter is the phrasebook itself, the subject's vocabulary, because the naming was part of the discovery. The movement's English, the skyglow, the glare, the trespass, the clutter; the Bortle grading, the limiting magnitude, the naked-eye count; the full cutoff, the warm spectrum, the lighting zone: the terms, coined and adopted across the literature, are the subject's own instruments of attention, the words that let a diffuse dissatisfaction become a discussable fact. The phrasebook's growth tracks the field's: the astronomers' terms first, the physics of the glow; the ecologists' and the physicians' next, the dose and the clock; the planners' last, the code's own technical prose, and the discovery of the subject, in the end, is visible in the language, a thing the era could talk about because it had first been given the words.

The discovery, in sum, is a story of attention: a gradual environmental change, too slow to make headlines and too ordinary to make anyone look up, was noticed, named, measured, and finally organized against, by the astronomers first and then by the biologists and the doctors and the citizens. The chapters that follow carry the story forward, from the discovery to the science, the measurement, and the movement, and the night's recovery, where it has come, is the story's closing lesson: the loss was slow, but the repair, where the will exists, is fast.